

Abdelrahman Abdelgawad

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Education

Master of Science (M.S.) in Computer Science

May 2028

Georgia Institute of Technology

Specialization in Computational Perception and Robotics

Bachelor of Science (B.S.) in Computer Science

May 2024

Pennsylvania State University

Minor in Mathematics, Minor in Cybersecurity Foundations

Experience

General Manager

Remote

Abdelgawad 1974 Inc.

November 2024 – Present

- Oversee day-to-day operations for 2 short-term rental cabins, managing 6+ guest stays per month and ensuring seamless check-in, check-out, and property readiness.
- Designed and implemented communication templates, improving response time and consistency across 100% of guest interactions.
- Coordinated logistics between guests, cleaning crews, and maintenance teams, resulting in 4.9+ average guest ratings across platforms.
- Managed operational expenses, tracked inventory, and ordered supplies, maintaining efficient budgeting and resource allocation for property upkeep.

Undergraduate Research Assistant

University Park, PA

Pennsylvania State University

January 2024 – May 2024

- Co-authored a research paper on training visual transformers, helping develop and clarify deep learning topics including CNNs, gradient descent, and image classification using MNIST and Fashion-MNIST datasets.
- Collaborated with faculty and TAs to write and revise 15+ textbook questions and sections, ensuring clarity, technical accuracy, and accessibility for students.
- Reviewed and formatted 20+ mathematical equations and theoretical explanations using LaTeX to support future versions of the course textbook.

Coding Camp Tutor

Brooklyn, NY

Girls Who Code

June 2023 – August 2023

- Co-led coding lessons for 20+ middle school students, introducing programming fundamentals (variables, loops, conditionals, and functions) through interactive Scratch projects.
- Assisted in designing and delivering 12 engaging lesson plans with instructors, fostering creativity and computational thinking.
- Provided one-on-one debugging and concept support, improving student understanding and confidence in coding.

Projects

[FencingTimeClone](#) (React, Google Cloud SQL)

- Developed a personal remake of fencingtimelive.com, a widely used platform for tracking and displaying live fencing tournament results.
- Designed a dynamic UI that allows users to add, modify, and delete fencers and events
- Programmed features for generating and presenting pools and direct elimination (DE) tables, incorporating match result input and real-time ranking updates.

[Playing Card Classifier](#) (Python, PyTorch)

- Constructed a deep learning model using PyTorch to classify playing cards with 96% accuracy.
- Automated the training and testing pipeline for a dataset of playing card images, refining model performance through hyperparameter tuning.
- Integrated visualization tools to showcase classification outcomes with clear, easy-to-understand charts and images, enhancing model interpretability.

Skills

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|-------------------------|-----------------------------|-------------------------------|
| • Java (Proficient) | • Linux (Intermediate) | • React (Familiar) |
| • Python (Proficient) | • Git (Intermediate) | • NoSQL (Familiar) |
| • SQL (Proficient) | • HTML, CSS (Intermediate) | • Microsoft Office Specialist |
| • C, C++ (Intermediate) | • JavaScript (Intermediate) | Master Certification (2019) |